

End-to-End Sales Analysis Project

Superstore Sales Analysis Using Python, MySQL, and Power BI

DATA ANALYTICS

BUSINESS INTELLIGENCE





Business Goal

Project Objective

Analyze historical sales data to identify the key drivers of **revenue and profitability** — from product performance to shipping efficiency.

Areas of Focus

The ultimate output: **actionable insights** to improve profitability and support data-driven decision-making.



Product & discount performance



Regional & segment behavior



Shipping efficiency & loss-making orders

Business Questions — Product & Discount

1

Product Analysis

- Which categories drive the highest sales and profit?
- Which sub-categories are most and least profitable?
- Which products show high sales but low profit?

2

Discount Analysis

- How do discounts affect overall profit?
- At what discount threshold does profitability begin to erode?



Business Questions — Region, Segment & Shipping

Regional Analysis

- Which regions generate the highest and lowest profit?
- Where is sales strong but profitability weak?

Customer Segment

- Which segments are the most profitable?
- Which segments receive the highest discounts?

Shipping Analysis

- How does shipping mode affect profit margins?

Order-Level Analysis

- Which individual orders generate the highest losses?

Tools & Technologies



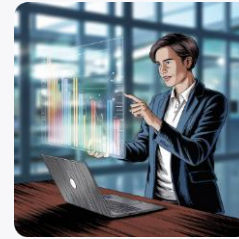
Python

Data cleaning and preprocessing using **Pandas** and **NumPy**



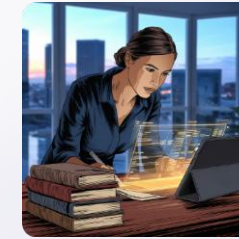
MySQL

Structured querying and relational data analysis



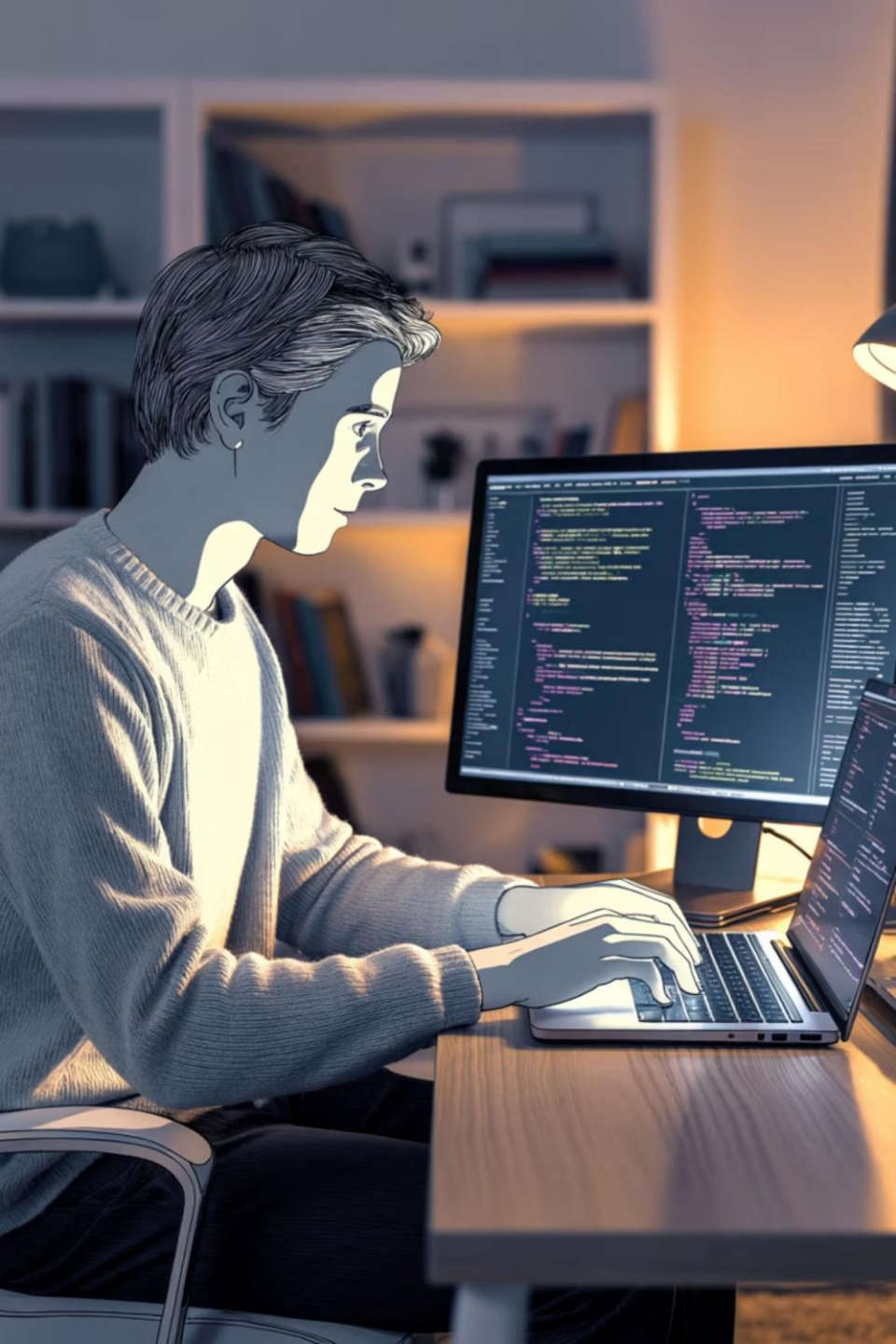
Power BI

Interactive dashboard design and visual storytelling



CSV Dataset

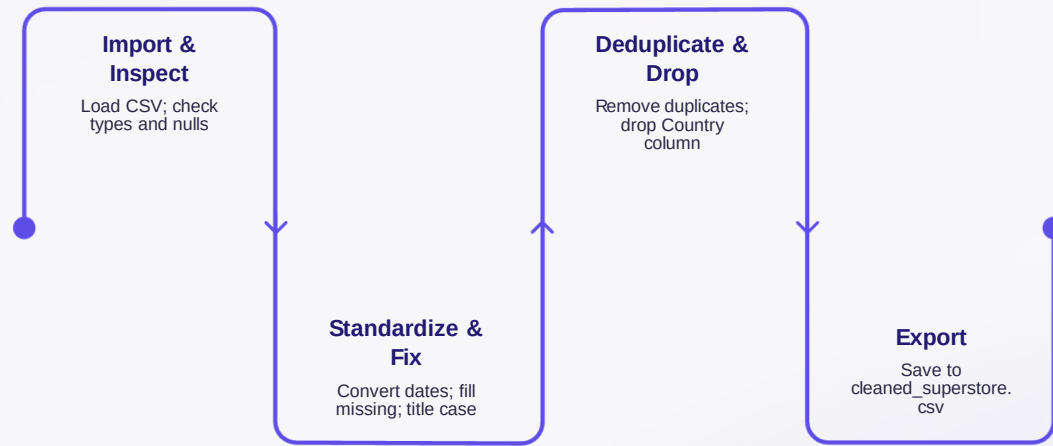
Sample Superstore dataset as the core data source



Python Data Cleaning & Preparation

The project began with rigorous data cleaning and preprocessing in Python — ensuring the dataset was **accurate, consistent, and analysis-ready** before loading into MySQL or Power BI.

Data Cleaning Steps



A structured, reproducible cleaning pipeline ensures data integrity at every stage before downstream analysis.

01

Imported dataset with Pandas; inspected types, nulls, and statistics

02

Converted Order & Ship Date columns to datetime format

03

Filled missing categoricals with *Unknown*; numerics via median imputation

04

Removed duplicates; standardized text (trim + title case)

05

Dropped redundant Country column; exported to cleaned_superstore.csv

Data Cleaning Results

0

Missing Values

Reduced to zero across all columns after imputation



Duplicates Removed

All redundant records eliminated for clean analysis

100%

Date Accuracy

All Order & Ship Dates converted to proper datetime format

1

Cleaned File

Final dataset exported as `cleaned_superstore.csv` for SQL and dashboard use

- ✓ The cleaned dataset is now fully prepared for structured SQL querying in MySQL and visual analysis in Power BI — ensuring all downstream insights are built on a reliable foundation.

MySQL Database Design and Analysis

The cleaned dataset was normalized into a relational database structure — enabling precise, scalable SQL-based analysis of sales, profitability, and customer behavior.

DATA ANALYTICS PROJECT



Database Schema

The schema separates descriptive attributes from transactional data, following a classic star schema pattern for analytical querying.



Dimension Tables

- Customers — segment, region, name
- Products — category, sub-category
- Locations — city, state, postal code
- Ship Modes — shipping method types

Fact Tables

- Orders — order date, customer, ship mode
- Order Items — sales, profit, discount, quantity



SQL Techniques Used



Joins

Linking dimension and fact tables across the schema



Aggregations

SUM, COUNT, AVG for group-level metrics



Window Functions

DENSE_RANK() for product and customer ranking



CASE / HAVING

Conditional logic and post-aggregation filtering

SQL Analysis: Product Performance & Discount Impact

Product Performance

Top Performers

Technology was the most profitable category. Copiers and Phones led sub-category profits.

Underperformers

Tables and Bookcases were the least profitable. High sales volume did not offset losses driven by heavy discounting.

Discount Impact



High Discounts

Profit Erosion

Profit declined consistently as discount rates increased. Excessive discounting was one of the strongest predictors of order-level losses.

Regional & Segment Performance



West Region

Generated the **highest total profit** of all regions. All regions demonstrated balanced performance — no region showed high sales paired with weak profitability.



Consumer Segment

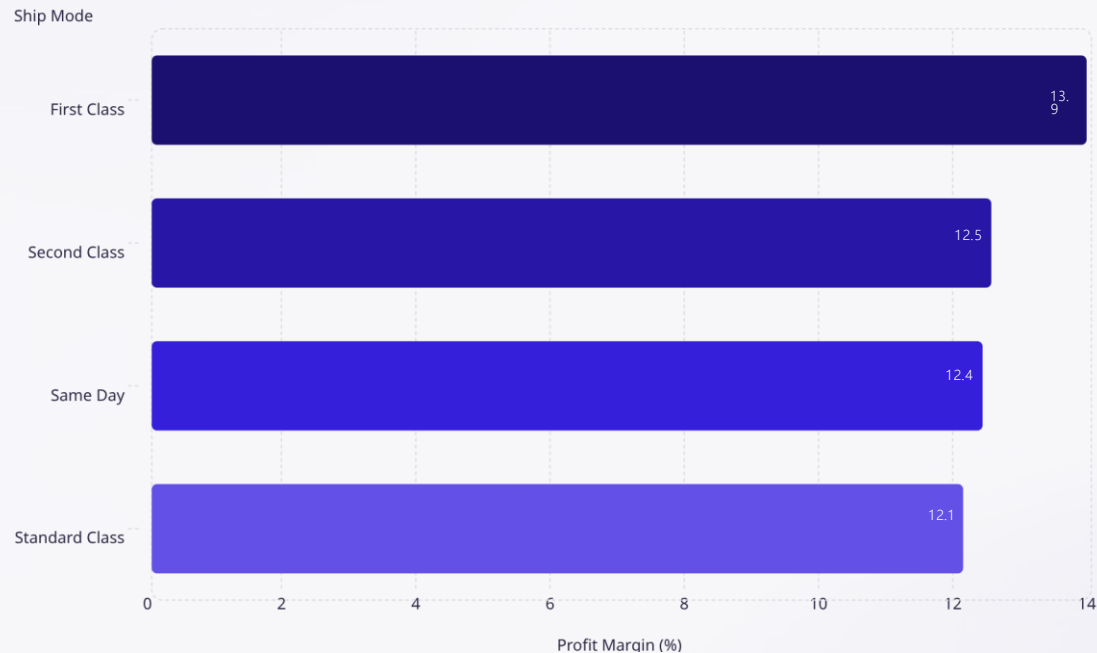
The **most profitable segment** and the top contributor to total sales volume — a consistent top performer across both metrics.



Order-Level Losses

Several **high-value orders produced significant losses**, confirming that large order size does not guarantee profitability without discount control.

Shipping Performance by Ship Mode



Key Takeaway

First Class shipping achieves the highest profit margin at **13.93%**, outperforming all other modes.

Margins across all ship modes are relatively consistent — ranging within ~2 percentage points — suggesting shipping method alone is not a major profitability driver.

i Standard Class, while the lowest margin at 12.09%, is likely the highest-volume mode — making overall contribution still significant.



Power BI Dashboard

An interactive one-page dashboard designed to visualize key business metrics, uncover trends, and support data-driven decisions across sales, profit, and operations.

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Dashboard Components

The dashboard is structured around four core KPIs and a flexible slicer panel — enabling dynamic filtering across time, geography, segment, and product category.

Total Sales

Overall revenue generated across all regions and segments

Total Profit

Net profit after costs and discounts

Total Orders

Volume of transactions processed

Profit Margin

Profitability ratio across all categories

Filters & Slicers

Interactive controls to slice data dynamically:

- Year — track performance over time
- Region — compare geographic performance
- Segment — analyze Consumer, Corporate, Home Office
- Category — drill into Technology, Furniture, Office Supplies

Visualizations at a Glance

Seven purpose-built visuals provide a comprehensive view of business performance — from macro trends to granular sub-category profitability.



Monthly Sales Trend

Tracks revenue fluctuations over time to identify seasonal patterns



Sales by Category

Compares contribution of Technology, Furniture, and Office Supplies



Profit by Region

Highlights geographic strengths and underperforming areas



Top 10 Sub-Categories

Ranks sub-categories by profitability to surface highest-value products



Discount vs. Profit

Reveals the margin impact of aggressive discounting strategies



Shipping Mode Margin

Evaluates profit margin across First Class, Second Class, and Standard

Key Business Insights: Revenue & Profit Drivers



What's Driving Performance

Technology was the strongest overall contributor to profit, with **Copiers** and **Phones** delivering the highest returns among all sub-categories.

The **Consumer** segment generated the most profit, making it the most commercially valuable audience across all regions.

- ✅ The **West** region was the top-performing geography — a benchmark for replicating success in other regions.

Profitability Risks & Operational Findings

Profitability Risks

Unprofitable Products

Tables and Bookcases were consistently loss-making — immediate review recommended

Discount Erosion

Excessive discounting significantly reduced margins across multiple categories

High-Value Losses

Several large individual orders produced major net losses — pricing controls needed

Operational Findings

First Class Shipping

Highest profit margin per order — premium service yields stronger returns

Standard Class Shipping

Highest sales volume overall, but at a structurally lower margin — efficiency trade-off to monitor



Margin compression from discounting and shipping mix requires strategic intervention to sustain profitability.

Project Deliverables

A structured, reproducible pipeline from raw data to executive dashboard — every artifact documented and ready for deployment.

1

Python Notebook

Data cleaning script and cleaned_superstore.csv output

2

MySQL Schema

Normalized relational database design with SQL analysis scripts

3

Power BI Dashboard

.pbix file with interactive visuals and dashboard screenshot

4

Final Report

Comprehensive project documentation with findings and recommendations

Skills Demonstrated



Data Engineering

Cleaning, preprocessing, and relational database design



Advanced Analytics

Exploratory data analysis and advanced SQL querying



Visualization & BI

Interactive dashboard development and insight generation

Project Outcome

Raw sales data transformed into a **complete, decision-ready BI solution** — surfacing profit drivers, quantifying discount damage, and exposing hidden order-level losses.

Profitability Map

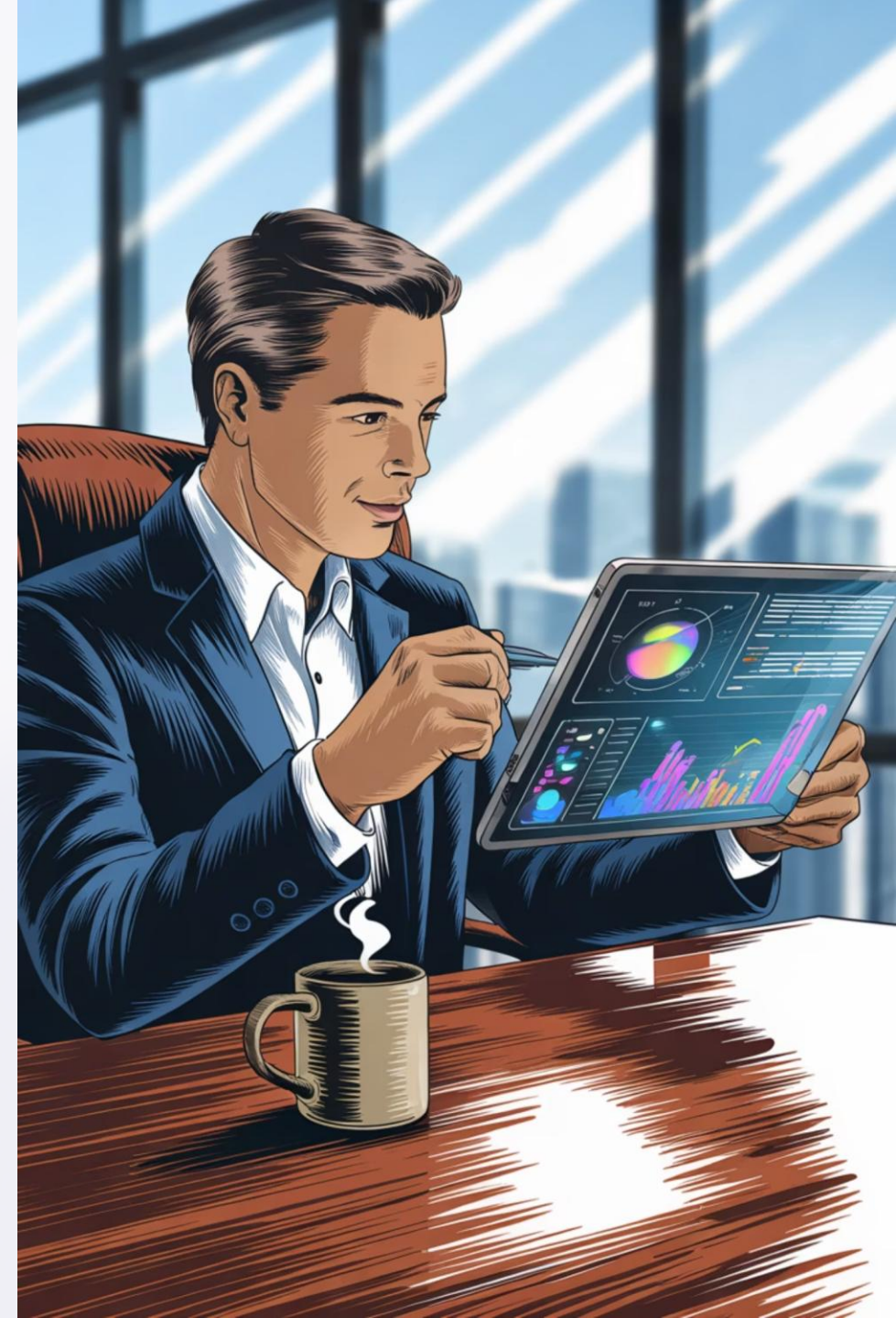
Identified top-performing products, regions, and customer segments

Discount Impact

Quantified how excessive discounting erodes margins across categories

Loss Detection

Uncovered hidden losses at the order level before they compound



Business Recommendations

Six high-impact actions to improve profitability and operational efficiency across the enterprise.



Reduce Discounting

Limit discounts on low-margin products — especially Tables and Bookcases



Focus on High-Margin Categories

Double down on Technology, Copiers, and Phones for maximum ROI



Audit Loss-Making Orders

Investigate and require approval for large negative-margin orders



Replicate West Region Success

Scale the strategies and pricing model driving West region outperformance



Optimize Shipping Modes

Align shipping decisions with profitability — avoid over-reliance on costly modes



Review Cost Structure

Conduct a full pricing and cost review for underperforming product lines